

East Crazies Example Soils Scope Of Work

DATE 2026-05-07	AUDIENCE WLG demonstration package	PURPOSE show that the system can produce resource-area scopes of work for a new NEPA project from a structured project intake
EXAMPLE PROJECT East Crazy Inspiration Divide land exchange	FOREST Custer Gallatin National Forest	DISTRICTS Yellowstone Ranger District; Bozeman Ranger District
RESOURCE AREA Soils	This is a planning and contracting support SOW for an early-stage project. It translates the controlling source stack into the fieldwork, analysis, and reporting needed for a concise soils resource report or supporting analysis in the proposal record. Project drivers: proposed trail construction or reroute, trailhead improvements, stream/wetland approaches, wet or hydric soils, erosion and sediment delivery, parcel restrictions, access/easement features, grazing-related improvements, and acquired-parcel management. This SOW is not a final agency action, agency NEPA finding, or decision document.	

DELIVERABLES

1. Fieldwork

- Data-needs, survey-trigger, and screen-out table for each proposed action element, identifying the source authority, reliable existing data used, required field survey if any, and reason any survey is not triggered.
- Activity-area, developed-site/permanent-road screen-out, and treatment-unit map set tied to soil map units, slope/stability limits, waterbody interfaces, access/staging, and crossing locations.
- DSD field package for each FW-STD-SOIL-01 activity area: pre-implementation percent DSD, expected post-implementation DSD, disturbance class, GPS/photo/form evidence, and protocol deviations.
- Site-specific soil/BMP observations for erosion, sediment delivery, wet or hydric soils, unstable areas, slopes over 40 percent, temporary access, blading, pits/trenches, burn piles, revegetation, and CWD only where FW-GDL-SOIL-07 or MON-SOIL-02 is triggered.

2. Analysis

- Potentially affected environment and reasonably foreseeable soil impacts for the proposed action, alternatives if any, and consequences of not implementing the action.
- FW-STD-SOIL-01 calculation package showing activity-area acreage, existing DSD, expected DSD, 15-percent standard status, restoration/net-improvement need, and any valid screen-out rationale.
- 36 CFR 219.15 Forest Plan consistency table for applicable soil desired conditions, standards, guidelines, suitability components if any, and MON-SOIL-01/MON-SOIL-02 monitoring triggers.
- Design criteria, mitigation, BMP, and monitoring matrix tied to action element, location, source trigger, timing, responsible party, verification method, and whether the measure is required or advisory.
- Data-gap and uncertainty summary identifying only the additional information essential to a reasoned choice among alternatives, the Forest Plan consistency conclusion, or the EA/FONSI support record.

3. Reporting

- Draft and final PDF Soils Resource Report or supporting analysis memo.
- Appendix A authority-to-report crosswalk updated for the project and cited in the report.
- Editable report source, final tables, map atlas, GIS package, photo log, field forms, source inventory, metadata/readme, and limitations memo.
- Concise EA/FONSI support text only where requested by the Forest Service and only from conclusions already supported in the soils report or supporting analysis memo.

FIELDWORK/DATA COLLECTION

Data collection will compile reliable existing soil information for the proposed project area and collect site-specific field information only where existing data are not sufficient to evaluate soil disturbance, stability, sensitivity, and productivity. Prioritize data gaps in areas with higher past disturbance, sensitive soil map units, wet or hydric soils, unstable slopes, erosion or sediment-connectivity risk, and areas expected to receive higher levels of ground disturbance from trail, trailhead, access, crossing, staging, restoration, or parcel-management work.

Before mobilization, prepare a data-needs and survey-trigger table that ties each proposed action element to the source requiring field verification. If reliable existing data are sufficient or a survey below is not triggered, record the reason in the table.

Methodology

Required data collection is based on current USDA NEPA procedures allowing use of reliable data sources and limiting new scientific or technical work to information essential to a reasoned choice, plus R1 FSM 2550 soil-quality direction, NFMA and 36 CFR 219.15 project consistency requirements, and the Custer Gallatin National Forest Land Management Plan. Use the Forest Service Soil-Disturbance Field Guide, Forest Soil Disturbance Monitoring Protocol, NRCS Web Soil Survey/SSURGO, and National BMP direction as the field and data methods.

- DSD surveys. Required where a management activity could create or add to detrimental soil disturbance in an activity area subject to FW-STD-SOIL-01 and reliable existing data are insufficient. Classify and estimate existing DSD and expected DSD for trail construction, trail reroutes, temporary access/staging, restoration,

crossing, and parcel-management work. Screen out intensively developed sites, permanent roads, and existing developed-recreation footprints from FW-STD-SOIL-01 calculations; address those areas through BMPs, site design, and watershed controls. Record compaction, rutting, displacement, puddling, surface erosion, severe burning if present, organic-matter loss, and mass movement. Tie applicable results to FW-STD-SOIL-01 and MON-SOIL-01.

- Soil survey and activity-area delineation. Required for each ground-disturbing action area where official soil data or plan consistency needs field verification. Verify soil map units, slope class, landform, drainage position, hydric/wet soil indicators, erosion hazard, revegetation limits, compaction/rutting limits, and sediment connectivity. Define the activity area used for DSD evaluation; for linear trails, document length, width, acreage, and rationale. If a developed-site or permanent-road footprint is screened out of FW-STD-SOIL-01, document the source rationale and the BMP/design controls that still apply.
- Soil sensitivity and productivity survey. Required where soil map units or field indicators show productivity risk, wetness, shallow or restrictive layers, erosion hazard, compaction sensitivity, poor revegetation potential, or other sensitive soil conditions. Identify design changes, restoration needs, or monitoring needed to maintain soil productivity and function.
- Slope, stability, and equipment-constraint survey. Required where mechanized construction, ground-based equipment, temporary access, blading, scarification, pits/trenches, burn piles, steep slopes, or unstable landforms are proposed. Verify sustained grades over 40 percent, high mass-wasting potential, unstable slopes, wet soils, and locations where FW-GDL-SOIL-01 through FW-GDL-SOIL-08 are triggered.
- CWD and organic-substrate survey. Required where vegetation management treatment units trigger FW-GDL-SOIL-07 or MON-SOIL-02. Estimate coarse woody debris where those components apply. Otherwise, document litter, duff, ground cover, and restoration substrate conditions only as existing-condition or BMP-support information.
- Waterbody interface and BMP survey. Required where trail, trailhead, access, crossing, drainage, or restoration work is near streams, wetlands, riparian areas, seeps, springs, swales, drainage paths, or other sediment-routing features. Identify drainage controls, hydrologic disconnection, crossing hardening, wet-weather restrictions, revegetation, decompaction, and inspection needs.

Minimum Survey Record

For each triggered survey location or activity area, record:

- date, observer, weather, soil moisture, access limits, survey trigger, and source driver;
- activity-area ID, action element, soil map unit, GPS identifier, photo IDs, and field form ID;
- protocol used, measurements taken, disturbance class, percent disturbance where required, and linear-feature substitute metric if approved;
- slope, landform, drainage position, hydric/wet soil indicators, unstable-area indicators, productivity or sensitivity indicators, and sediment-connectivity notes;
- BMP, design criterion, mitigation, restoration, monitoring requirement, data gap, or revisit need.

Conduct fieldwork during snow-free conditions when the mineral soil surface, organic cover, and drainage paths can be observed. Conduct a wet-season or post-storm revisit only if waterbody connectivity, runoff, erosion, or sediment delivery cannot be determined during the main survey.

ANALYSIS OUTPUT

Use Appendix A as the authority-to-report control table. The analysis should answer only the substantive soil issues and plan-consistency questions triggered by the proposed action and should tie each conclusion to reliable existing data, field evidence where needed, GIS data, cited source direction, or a documented Forest Service assumption.

1. Existing Condition

Summarize the soil setting for each affected activity area:

- soil map unit, slope, landform, drainage, hydric/wet soil indicator, erosion hazard, and revegetation or compaction limitation;
- existing detrimental soil disturbance, route drainage, erosion, sediment connectivity, and restoration need;
- CWD, litter/duff, ground cover, or organic-substrate condition only where the CWD/organic-substrate survey is triggered;
- disposal-parcel and acquisition-parcel soil conditions only where parcel restrictions or future NFS management create a soil issue;
- data gaps that limit effects conclusions.

2. Effects Indicators

Use the same indicators for the proposed action, alternatives if any, and consequences of not implementing the action:

- activity-area acreage for area-based work;
- length, width, crossing count, drainage feature count, wetland/riparian approach count, and representative disturbance class for linear trail work;
- percent detrimental soil disturbance where FW-STD-SOIL-01 applies;
- hydrologic connectivity, erosion risk, revegetation feasibility, wet-soil limitation, unstable-area risk, and restoration need.

Do not force linear trail effects into acreage-only conclusions. Use acreage where it is useful and linear metrics where they better describe the effect.

3. Proposed-Action Effects

Analyze reasonably foreseeable soil impacts from the proposed trail construction, trail reroute, trailhead work, crossings, access/staging, restoration, and parcel-management changes. Address whether the action would:

- create, add to, or restore detrimental soil disturbance;
- route sediment toward streams, wetlands, riparian areas, seeps, springs, swales, or drainage paths;
- reduce an existing stream/wetland crossing effect through the Inspiration Divide Trail reroute;
- require erosion, compaction, drainage, topsoil, revegetation, wet-weather, or equipment controls;
- create soil-management obligations on acquired NFS parcels or reduce soil risk through parcel restrictions after conveyance.

4. Forest Plan Consistency

Use Appendix A to document the Forest Plan consistency finding needed for the soils report. The report must state:

- whether FW-STD-SOIL-01 applies, is met, requires restoration, or is not triggered;
- which FW-GDL-SOIL-01 through FW-GDL-SOIL-08 guidelines apply and what design feature or equivalent measure satisfies each one;
- whether MON-SOIL-01 or MON-SOIL-02 monitoring is required or screened out;
- which related road/trail, watershed, riparian, wetland, revegetation, invasive-species, backcountry, or recommended-wilderness components change the soils analysis.

5. Mitigation, BMPs, And Monitoring

Provide one mitigation/BMP/monitoring matrix. Each row must include action element, location, triggering source, soil concern, required measure, timing, responsible party, and verification method. Include only measures tied to a triggered source or a site-specific field finding.

6. Reasonably Foreseeable Impacts And Uncertainty

Evaluate only the reasonably foreseeable soil impacts of the proposed action and alternatives, if any, using the indicators listed above. Include separate-action context only where the Forest Service determines it is necessary to inform the NEPA finding or an applicable soil plan-consistency conclusion.

Close the analysis with unresolved assumptions, field limitations, missing design details, and any information that limits use of the report in the proposal record.

REPORTING OUTPUT

1. Draft Soils Resource Report Or Supporting Analysis Memo

Provide a draft report or supporting analysis memo with:

- project action elements and soils issue triggers;
- field methods and survey-trigger table;
- existing condition by activity area;
- effects indicators and disturbance calculations;
- reasonably foreseeable environmental impacts where triggered, including separate-action context only where the Forest Service determines it is necessary;
- Forest Plan consistency finding using Appendix A;
- mitigation/BMP/monitoring matrix;
- residual uncertainty, assumptions, and limitations.

Acceptance criteria:

- directly addresses each soil-triggering action element;
- states whether FW-STD-SOIL-01 applies, is met, requires restoration, or is not triggered;
- ties conclusions to Appendix A, field evidence, GIS evidence, official soil data, or specialist judgment;
- distinguishes required mitigation or monitoring from optional design recommendations.

2. EA/FONSI Support Text

If requested by the Forest Service, provide concise disclosure text drawn from the soils report or supporting analysis memo.

Acceptance criteria:

- summarizes the potentially affected environment, reasonably foreseeable environmental impacts, mitigation/design criteria, and Forest Plan consistency only where supported by the report;
- references the soils report or supporting analysis memo instead of duplicating technical detail;
- identifies design criteria or monitoring that support the EA/FONSI analysis.

3. Final Report And Data Package

Provide final report or memo, final PDF, editable source file, Appendix A, final tables, map atlas, GIS layers or geodatabase/geopackage, photo log, field forms, source data inventory, metadata/readme, and limitations memo.

Acceptance criteria:

- preserves source citations, field-data traceability, and the authority-to-report crosswalk;
- documents residual uncertainty and limits on use of the analysis;
- includes the 7 CFR section 1b.10 financial-interest disclosure statement if the Forest Service authorizes third-party preparation of an EA or EIS, in whole or in part.

APPENDIX A. AUTHORITY-TO-REPORT CROSSWALK

Use this appendix as the starting control table for the soils resource report. Update it only if current regulations, applicable soil guidance, source currentness, or the proposed action changes.

A1. NEPA And USDA NEPA Procedures

SOURCE	NEPA; USDA NEPA procedures at 7 CFR part 1b.
TRIGGER	Federal land exchange, trail, trailhead, access, and parcel-management decisions that need soils information for an environmental assessment, finding of no significant impact, or proposal record.
REPORT OUTPUT	concise affected environment, methods, reasonably foreseeable environmental impacts, alternatives if any, consequences of not implementing the action, mitigation/monitoring, source citations, and data limitations.
ACCEPTANCE TEST	each conclusion is traceable to reliable existing data, field evidence, or a documented Forest Service assumption; the report does not make the agency decision.

A2. Third-Party Or Consultant-Prepared Analysis

SOURCE	7 CFR section 1b.10 applicant/third-party preparation procedures.
TRIGGER	WLG or a consultant prepares environmental information or soils analysis for Forest Service review under Forest Service supervision.
REPORT OUTPUT	editable report, field forms, GIS/map package, photo log, source inventory, assumptions, limitations, and disclosure statement if required by 7 CFR section 1b.10.
ACCEPTANCE TEST	Forest Service can independently review the methods, source citations, field evidence, and assumptions before using the report.

A3. NFMA And 36 CFR Part 219

SOURCE	NFMA; 36 CFR part 219, especially project/activity consistency with the applicable Forest Plan.
TRIGGER	Any NFS project or authorization after approval of the Custer Gallatin Forest Plan.
REPORT OUTPUT	Forest Plan consistency section that addresses applicable desired conditions, standards, guidelines, suitability, and monitoring components.
ACCEPTANCE TEST	the report states whether the action complies with standards, follows guidelines or uses an equally effective design, and avoids foreclosing applicable desired conditions.

A4. Custer Gallatin Forest Plan Soil Desired Conditions

SOURCE	FW-DC-SOIL-01, FW-DC-SOIL-02, and FW-DC-SOIL-03.
TRIGGER	Trail, trailhead, crossing, access, restoration, and parcel-management actions that could affect soil productivity, hydrologic function, erosion, or ground cover.
REPORT OUTPUT	existing-condition and effects discussion for soil productivity, erosion, sediment connectivity, infiltration, ground cover, and restoration feasibility.
ACCEPTANCE TEST	the report explains how the action maintains or moves toward the soil desired conditions, or identifies the design changes needed.

A5. Custer Gallatin Forest Plan Soil Standard

SOURCE	FW-STD-SOIL-01.
TRIGGER	Any management activity that could create or add to detrimental soil conditions in an activity area subject to the standard. Screen out intensively developed sites, developed recreation sites, administrative sites, rock quarries, mines, and permanent roads where the standard does not apply.
REPORT OUTPUT	activity-area delineation, existing DSD estimate, expected DSD estimate, linear-trail metric rationale if used, screen-out rationale where used, and restoration/design measures needed to avoid exceeding the standard.
ACCEPTANCE TEST	the report states whether FW-STD-SOIL-01 applies, is met, requires restoration, or is not triggered.

A6. Custer Gallatin Forest Plan Soil Guidelines

SOURCE	FW-GDL-SOIL-01 through FW-GDL-SOIL-08.
TRIGGER	Ground-based equipment, temporary access, blading, scarification, pits/trenches, burn piles, steep slopes, high mass-wasting potential, CWD, or restoration work.
REPORT OUTPUT	guideline applicability table with action element, location, field finding, design feature, mitigation, or equally effective alternative.
ACCEPTANCE TEST	every triggered guideline is carried into the design criteria or explicitly screened out.

A7. Custer Gallatin Forest Plan Soil Monitoring

SOURCE	MON-SOIL-01 and MON-SOIL-02.
TRIGGER	DSD monitoring where FW-STD-SOIL-01 applies; CWD monitoring where vegetation management treatment units trigger FW-GDL-SOIL-07 or MON-SOIL-02.
REPORT OUTPUT	monitoring recommendation with location, metric, timing, responsible party, and success criterion.
ACCEPTANCE TEST	the report identifies pre/post DSD monitoring and CWD monitoring where triggered, or states why they are not triggered.

A8. Region 1 Soil Management Direction

SOURCE	USDA Forest Service Manual, Region 1 Supplement, FSM 2550 Soil Management.
TRIGGER	Forest Service management activity that could affect soil productivity or create detrimental soil disturbance.
REPORT OUTPUT	regional soil-quality method, DSD threshold logic, productivity-risk discussion, and restoration requirements.
ACCEPTANCE TEST	regional soil direction is visible in the methods, effects indicators, and mitigation/monitoring matrix.

A9. Soil-Disturbance Field Guide And Monitoring Protocol

SOURCE	Forest Service Soil-Disturbance Field Guide and Forest Soil Disturbance Monitoring Protocol.
TRIGGER	Any required DSD survey or soil-disturbance classification.
REPORT OUTPUT	disturbance classes, field forms, photo/GPS evidence, activity-area summary, and protocol deviations.
ACCEPTANCE TEST	disturbance conclusions are repeatable from the field record.

A10. NRCS Web Soil Survey And SSURGO

SOURCE	NRCS Web Soil Survey and SSURGO.
TRIGGER	Every mapped activity area and any parcel where soil limitations affect the proposed action.
REPORT OUTPUT	soil map unit/activity-area crosswalk, limitation ratings, hydric/wet soil indicators, erosion hazard, drainage, revegetation limits, and data limitations.
ACCEPTANCE TEST	official soil data are cited and field-verified where the action creates a soil issue.

A11. National BMPs

SOURCE	Forest Service National Best Management Practices Program.
TRIGGER	Trail, trailhead, access, crossing, drainage, or restoration work with water-quality or sediment-delivery risk.
REPORT OUTPUT	BMP matrix for drainage control, hydrologic disconnection, crossing hardening, sediment control, wet-weather restrictions, decompaction, revegetation, and inspection.
ACCEPTANCE TEST	BMPs are tied to mapped locations and verification methods, not listed generically.

A12. East Crazies Example Action Elements

SOURCE	East Crazies structured intake assumptions and Forest Service project summary for project 63115.
TRIGGER	Sweet Trunk Trail work, Inspiration Divide Trail reroute, Big Timber Canyon Trailhead improvements, stream/wetland approaches, access/easement features, parcel restrictions, grazing improvements, and acquired-parcel management needs.
REPORT OUTPUT	action-element by soil-concern matrix, field-survey trigger table, effects indicators, maps, mitigation/BMP/monitoring matrix, and concise rationale for any non-substantive soil issue.
ACCEPTANCE TEST	every soils conclusion is tied to an action element or explicitly screened out.

SOURCES CITED

- National Environmental Policy Act, 42 U.S.C. 4321 et seq.
- National Forest Management Act, 16 U.S.C. 1600 et seq.

- USDA NEPA procedures, 7 CFR part 1b: <https://www.ecfr.gov/current/title-7/subtitle-A/part-1b>
- National Forest System land management planning, 36 CFR part 219: <https://www.ecfr.gov/current/title-36/chapter-II/part-219>
- USDA Forest Service Manual, Region 1 Supplement, FSM 2550 Soil Management: <https://www.fs.usda.gov/im/directives/field/r1/fsm/2500/r1-2550.docx>
- Custer Gallatin National Forest Land Management Plan: <https://www.fs.usda.gov/r01/custergallatin/planning/forest-plan/custer-gallatin-land-management-plan-forest-plan-revision>
- East Crazy Inspiration Divide Land Exchange project summary: <https://www.fs.usda.gov/r01/custergallatin/projects/63115>
- Forest Service Soil Quality Monitoring resources: <https://www.fs.usda.gov/soils/monitoring.shtml>
- Forest Service Soil-Disturbance Field Guide: https://www.fs.usda.gov/t-d/php/library_card.php?p_num=0819+1815P
https://www.fs.usda.gov/rm/pubs_other/rmrs_2009_napper_c001.pdf
- NRCS Web Soil Survey and SSURGO: <https://www.nrcs.usda.gov/resources/data-and-reports/web-soil-survey>
- Forest Service National Best Management Practices Program: <https://www.fs.usda.gov/naturalresources/watershed/bmp.shtml>
- East Crazies example project intake assumptions: land exchange, proposed trail construction, proposed trail reroute, trailhead improvements, parcel restrictions, easements, wetland/riparian interfaces, grazing-related improvements, and acquired-parcel management needs.